

CS306 Spring 2025-2026 - Project Phase 1

1. Form your groups. The maximum group size is three students.

Use the following Google Sheet to indicate your group members:

https://docs.google.com/spreadsheets/d/1GVBf8oiYtA5T7Fg4bn-ij49MZvVZeL6DZVcWPZIT_c/edit?gid=0#gid=0

2. Decide on a database application and write a one-page description explaining what you plan to develop.

Based on your description, draw the ER diagram which will include all constraints such as key constraints, participation constraints, etc that you specified in your description. .

Submit your work as a PDF through Sucourse which should include 2 sections and an Appendix:

(1) Project title and description

Your project title should describe the scope of your application. For example: University Database Application for Students.

Project description should be detailed enough to describe the data that will be stored in the database. The readers of your description should understand the entities, their attributes and relationships among entities. Your description will also include constraints that we have seen in class such as keys, participation constraints as well as the constraints on the type of the relationships.

Project description should be written in a way that allows a correct Entity–Relationship diagram (using the original Chen notation) to be constructed directly from the text.

(2) ER Diagram (Model)

Each group should submit **only one report**. One student from each group should handle the submission. Please include the **names and student IDs of all contributing group members** on the first page of the PDF.

Q: *How large should the database be in terms of the number of tables and data input?*

A: The formula is at least $2n+2$ entities in your ER diagram where n is the group size. For example, if there are 2 students in the group, you are expected to have at least 6 entities. We prefer not to put a lower bound on the number of relationships in your ER diagram.

(3) Appendix — AI Usage Documentation

You are free to use (and even encouraged to use) an AI tool to assist you in your project. Students must document:

- If they used an AI tool for this step of the project and if so which AI tools were used (ChatGPT, Gemini etc)
- Exact prompts used to form the project description and the ER diagram
- Full AI responses (or summarized but complete transcripts)
- Which parts were directly used, modified, or rejected
- How the final ER diagram differs from AI suggestions

Let me remind you that we have covered the Chen ER Model Version in class. This is the original ER model notation that was proposed in the original paper by Peter Chen with minor modifications. These modifications include representing 1 to Many type of relationships with an arrow for ease of drawing by tools and for pedagogical reasons. **Make sure that your final ER diagram matches what you learned in class.**